When Structure Doesn't Matter: Loss-Invariant Routing Specialisation in Hybrid Language Models

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Abstract

We propose two parameter-efficient architectural additions to hybrid state-space/attention language models — *hormone routing*, a token-conditional router that injects a gated combination of frozen direction vectors into the residual stream after each attention block, and *intracellular attention*, a slot-attention mechanism over the d_{state} memory cells of a selective state-space block — and use them to investigate the relationship between architectural inductive bias and the optimisation objective. Both mechanisms use a zero-initialised output gate so that the unmodified hybrid baseline is recovered exactly at initialisation. We pretrain a 125M-parameter Mamba/MQA hybrid on FineWeb-Edu under a fixed 2B-token budget and report a striking decoupling between the routing module's representational organisation and its effect on the loss objective. The hormone-routing module develops semantically-coherent per-genre routing specialisation across every variant we examine — including controls that replace the extracted directions with random unit vectors, with vectors extracted from a randomly-initialised model, or that freeze the output gate at 1.0. Maximum pairwise L_1 distance between genre routing distributions ranges from 0.6 to 1.6 (uniform routing $= 0.0$, disjoint $= 2.0$), with the variant of *highest* specialisation strength being the one whose gate the model cannot modulate. Yet **specialisation strength does not correlate with loss**: the four properly-initialised variants converge to identical validation perplexity (26.13). To probe whether this loss-invariance reflects a fundamental ceiling on the mechanism's contribution or a regime the model actively absorbs, we train a sixth variant with a forced non-zero gate initialisation and a non-trainable $3\times$ amplification on the injection. The model responds by drifting the gates downward and shrinking the per-hormone magnitudes, but the residual perturbation it cannot suppress — a roughly $2\times$ amplification of the unforced regime — does not damage validation loss at the same training point. The picture that emerges is of a *homeostatic envelope*: within it, the gated injection is absorbed without affecting predictions, and the routing module's representational organisation develops freely. The envelope is wider than the unforced variants naturally select, and the model expresses a preference for smaller perturbations within it without requiring them. The envelope can still be broken — by introducing the routing during base pretraining rather than as a continuation from a stabilised checkpoint — which damages the loss by $+2.07$ PPL even though the same routing module develops similar specialisation in this regime as in any other. The loss-relevant axis is the training schedule, not the perturbation magnitude. Both contributions add fewer than 1% to the base model's parameter count and are released as open-source code.

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1 Introduction

Hybrid architectures that interleave selective state-space mixers [11] with attention [36] have emerged as a strong alternative to pure-attention transformers, combining the linear-time long-range modelling of structured state-space models (SSMs) with the precise content-based retrieval that attention provides [23, 10, 28]. These hybrids retain attention's expressivity at the locations where it matters most, while relegating the bulk of the sequence mixing to the more parameter-efficient SSM path.

We propose two parameter-efficient architectural additions to this family and use them to investigate the relationship between architectural inductive bias and the optimisation objective. The two additions target distinct loci. *Intracellular attention* concerns the SSM's *internal* structure: the selective state $h_t \in \mathbb{R}^{d_{\text{inner}} \times d_{\text{state}}}$ of a Mamba block is best read as d_{inner} parallel channels, each carrying a d_{state}-dimensional memory; within each channel the slots evolve independently and only ever interact through the linear readout $y_t = h_t C_t^\top$. We replace this linear contraction with full content-dependent attention over the slots. *Hormone routing* concerns the residual stream *between* blocks: it injects a token-conditional combination of a small bank of frozen direction vectors — extracted from the partially-trained base model itself via document-style contrast pairs [35, 30, 39] — after each attention block, gated by a learned scalar.

Both mechanisms share a design discipline. Each adds fewer than 1% to the base parameter count. Each is gated such that, at initialisation, the modification is an exact no-op and the hybrid backbone is recovered bit-for-bit. Each is structurally orthogonal to the other.

Our empirical contribution is qualitatively different from the contribution we initially set out to make. Across every hormone-routing variant we tested at the 125M-parameter, 2B-token scale — with extracted directions, with random unit vectors, with the output gate frozen at 1.0, and with directions extracted from a randomly-initialised model — the router converges to a strongly genre-discriminative routing distribution (max pairwise L_1 distance between routing distributions across six text genres: 0.6–1.6, vs 0.0 for uniform routing). The specialisation is qualitatively interpretable in the variants where the directions carry semantic labels (in the extracted-direction variant, news routes preferentially to the urgency hormone, fiction to curiosity, code to skepticism, urgent text to warmth) and is structurally analogous in the variants where they do not. Yet none of these variants moves the validation perplexity needle: all properly-initialised hormone variants achieve identical validation PPL (26.13), matching the unmodified base. Only the variant that introduces hormone routing during base pretraining — rather than as a continuation from a stabilised checkpoint — changes the loss, and it does so by degrading it (28.20 PPL).

We interpret this finding as an instance of a broader phenomenon: **the model's architectural channels can develop structured representational organisation that is not visible in the loss**. The router learns the genre-conditional distribution it would need in order to use the hormone bank meaningfully, but the bounded perturbation magnitude — gate g converges to 0.23 at the upper attention layer, ≈ 0 at the lower one — is too small to alter the model's predictions at this scale. Whether this represents a fundamental ceiling on the mechanism's contribution, or merely a capacity-bottlenecked regime in which the residual stream is too low-dimensional to carry useful gradient signal, is a question we partially address with a forced-engagement variant (Section 6).

Our contributions are as follows.

- **Empirical: a clean decoupling between routing specialisation and loss.** Across five hormone-routing variants pretrained on a 125M-parameter Mamba/MQA hybrid for 2B tokens, the routing module develops strongly genre-discriminative routing distributions in every condition we test — including controls that replace the extracted directions with random unit vectors and that freeze the gate at 1.0 — with maximum pairwise L_1 distance between genre routing distributions in the range 0.6–1.6 (uniform $= 0.0$, disjoint $= 2.0$). Yet the variant with the *highest* specialisation strength achieves the same validation perplexity as the variant with the *lowest* and as the unmodified baseline. **Specialisation strength does not predict downstream loss**, and the model is invariant to the routing module's content (extracted vs random vs fixed-gate) but not to its training-schedule timing (continuation vs from-scratch).
- **Architectural: hormone routing.** The mechanism that produces the above finding is a parameter-efficient routing channel: a per-token softmax-gated linear combination of a small

bank of frozen direction vectors, added to the residual stream after each attention block, multiplied by a learned scalar gate. The router is small (under 0.01% of the base model's parameters) and the bank is non-trainable. We further propose a *document-style contrast-pair extraction protocol* that recovers behavioural directions from the base model itself — without instruction-tuning data or annotated preferences — and discuss the methodological reasons for preferring it to instruction-style pairs in the pretraining setting.

- **Architectural: intracellular attention.** A second proposed mechanism, structurally orthogonal to hormone routing, places content-dependent attention inside the SSM scan to let the d_{state} state slots attend to each other before the linear readout. The mechanism preserves the SSM's recurrent form, adds parameters proportional only to a small bottleneck dimension $d_{\text{attn}} \ll d_{\text{inner}}$, and is gated identically (zero-init, learnable) so that the unmodified scan is recovered at initialisation.
- **Forced-engagement test.** A sixth variant initialises the gate at a non-zero value and amplifies the injection by a constant multiplier, producing an effective initial perturbation several times larger than the largest converged gate of the unforced variants. This isolates whether the loss-decoupling reflects a fundamental property of the mechanism or a capacity-bottlenecked regime in which the model can absorb only small perturbations.

The rest of the paper is structured as follows. Section 2 situates the work against the SSM, hybrid, representation-engineering, and routing literatures. Section 3 gives the hybrid backbone. Sections 4 and 5 present the two architectural contributions in turn. Section 6 reports the empirical study. Section 7 characterises what the router learns and discusses the implications for representation-level organisation independent of objective.

2 Related Work

State-space models for sequence modelling. Structured state-space models [12, 14] reformulate sequence mixing as a parameterised linear dynamical system, recovering near-linear sequence-length scaling. The S4 family [14] and the diagonal variants [15, 13] introduced practical kernels and parameterisations. Mamba [11] added input-dependent selectivity: the discretisation step Δ_t, the input matrix B_t, and the readout C_t are all functions of the current input, granting the SSM a content-based memory mechanism. Mamba-2 [7] unifies the SSM update with a structured-attention view and improves hardware utilisation. We build on Mamba-1 and treat its state h_t, of shape $d_{\text{inner}} \times d_{\text{state}}$, as the natural locus of the intracellular-attention mechanism we propose.

Hybrid SSM–attention architectures. Pure-SSM models lag pure-attention transformers on tasks requiring precise in-context recall [19, 2]. Hybrid models address this by interleaving a small number of attention layers among many SSM layers. Jamba [23] popularised the design at scale; Zamba [10] explored parameter-sharing across the attention layers; Park et al. [28] systematically studied layer-mixing ratios and found that even a single attention layer per block of SSM layers recovers most of the lost expressivity. We adopt a 5:1 SSM-to-attention ratio with multi-query attention [31] and rotary positional embeddings [34] for the attention layers. Our two contributions are orthogonal to the choice of mixing ratio and could be applied to any of the above hybrid designs.

Activation steering and representation engineering. A growing body of work shows that LLM behaviour is modulated by a small number of linear directions in the residual stream. Zou et al. [39] introduced representation-engineering as a framework for reading and writing such directions. Turner et al. [35] and Rimsky et al. [30] demonstrate that adding a difference-of-means vector — computed from contrastive prompts — to the residual stream at inference time can steer attributes like truthfulness, refusal, and persona. Earlier work on contrast-consistent search [5] and probing [1] likewise establishes that interpretable axes are present in unmodified models. All of this work, however, applies the directions *post-hoc* at inference time, with weights chosen by the user. Our hormone-routing module instead re-injects extracted directions *during pretraining* through a learned router that decides per-token how to combine them; the directions themselves are kept frozen so that their geometric meaning is preserved through training.

Routing and mixture-of-experts. Mixture-of-experts (MoE) models [33, 9, 20] use learned routers to dispatch tokens to specialised expert subnetworks. Our hormone router is structurally analogous —

a linear-then-softmax map from the hidden state to a distribution over a small, fixed set of directions — but the "experts" here are not parameter banks; they are *frozen direction vectors* of dimension d_{model}. The routing decision selects which behavioural axis to attend to, not which subnetwork to compute. This permits a richer routing signal at a vanishing parameter cost.

Soft prompts and prefix tuning. Trained prompt embeddings [21, 22] introduce a small number of trainable vectors prepended to the input; these are conceptually related to our hormones in that both use a small bank of d_{model}-dimensional vectors to modulate behaviour. The differences are substantive: hormone vectors are not prefix tokens, are not exposed to attention, are token-conditional rather than fixed across the sequence, and are extracted from the base model rather than initialised randomly.

Slot-based attention. "Slots" have appeared in the visual representation-learning literature [24, 18] as latent tokens that summarise scene content via attention. Our use of the term refers to the d_{state} memory cells inside an SSM state and is unrelated; the conceptual link is only that both designs apply attention over a small, learned set of summary vectors. To our knowledge, no prior work applies attention over the state dimension of a selective SSM.

Probing and the structure-vs-task gap. A long line of work probes neural-network internals for learned structure that is not obviously demanded by the training objective [1, 16, 3]. Much of this literature implicitly treats discovered structure as evidence of the network's competence on related tasks: a probing classifier that recovers part-of-speech tags from intermediate representations, for example, is taken to demonstrate that the network "knows" something useful about syntax. Hewitt and Liang [16] push back on this convention by showing that high-capacity probes can recover arbitrary structure regardless of whether the network exploits it. Our routing-specialisation finding is in the same spirit but at a different locus: the routing module is not a post-hoc probe but a built-in architectural channel, and the structure it develops is not a property we infer from a trained classifier but a property of the model's own forward pass. The decoupling we observe between specialisation strength and downstream loss is a stronger claim than "the probe finds structure that does not matter"; it is "the model itself produces and uses structure that does not matter for its training objective," at least at the scale we examine.

3 Architecture

We use a hybrid Mamba/MQA backbone of L residual layers with a fixed 5:1 SSM-to-attention layer pattern (i.e. five SSM layers between every attention layer, with a final attention layer if $L \bmod 6 \neq 0$). All layers share the same hidden width d_{model}. Layer norms are RMSNorm [38]; word embeddings and the language-model head share weights.

SSM layers. Each SSM layer applies pre-norm and a Mamba-1 selective SSM block [11], with no separate channel-mixing MLP. Concretely, given the layer input $x \in \mathbb{R}^{B\times T\times d_{\text{model}}}$, we project to twice the inner width, $[x', z'] = W_{\text{in}}\,\text{RMSNorm}(x)$ with $x', z' \in \mathbb{R}^{B\times T\times d_{\text{inner}}}$ and $d_{\text{inner}} = 2d_{\text{model}}$, apply a causal depthwise convolution to x', and run the input through a selective scan parameterised by an input-dependent discretisation step Δ_t, input matrix $B_t \in \mathbb{R}^{d_{\text{state}}}$, and readout matrix $C_t \in \mathbb{R}^{d_{\text{state}}}$. The state $A \in \mathbb{R}^{d_{\text{inner}}\times d_{\text{state}}}$ is parameterised in log-space as a per-channel diagonal. The scan output is gated multiplicatively by $\text{SiLU}(z')$ and projected back to width d_{model}:

$$h_t = \Delta_t\, A \odot h_{t-1} + \Delta_t\, B_t \odot x_t, \tag{1}$$

$$y_t = h_t\, C_t^{\top} + D \odot x_t, \tag{2}$$

with $h_t \in \mathbb{R}^{d_{\text{inner}}\times d_{\text{state}}}$ and $D \in \mathbb{R}^{d_{\text{inner}}}$ a per-channel skip. Section 4 replaces the linear contraction $h_t C_t^{\top}$ with attention over the d_{state} slots.

Attention layers. Each attention layer applies pre-norm followed by multi-query attention [31] with a single shared key/value head, rotary positional embeddings on the query and key projections, and a SwiGLU [32] channel-mixing MLP of inner width $d_{\text{ff}} = \lceil {}^{8}\!/_{3} d_{\text{model}} \rceil$ rounded to a multiple of

Table 1: Backbone configurations. All sizes share $d_{\text{state}} = 16$, $d_{\text{attn}} = 16$, head dim 128, MQA, 5:1 SSM:Attn pattern.

Name	Params	d_{model}	L	n_{heads}	SSM/Attn layers
TINY	$\sim$125M	1024	12	8	10/2
SMALL	$\sim$350M	1536	18	12	15/3
MEDIUM	$\sim$770M	2048	24	16	20/4

64. The hormone routing layer of Section 5 adds a third residual addition after the MLP residual:

$$x \leftarrow x + \text{Attn}(\text{RMSNorm}(x)), \tag{3}$$

$$x \leftarrow x + \text{MLP}(\text{RMSNorm}(x)), \tag{4}$$

$$x \leftarrow x + \text{Hormone}(x). \tag{5}$$

Configuration ranges. All experiments in this paper use the TINY configuration ($d_{\text{model}} = 1024$, $L = 12$, $d_{\text{inner}} = 2048$, $d_{\text{state}} = 16$, eight query heads, one shared KV head, head dim 128, ≈ 125M parameters), which admits full ablation within a 2B-token budget on consumer-grade hardware. A 350M-parameter SMALL and a 770M-parameter MEDIUM variant are also defined for future scaling experiments. Layer counts and widths for each configuration are summarised in Table 1.

4 Intracellular Attention

Motivation. Inside a Mamba-1 selective SSM block, the recurrent state $h_t \in \mathbb{R}^{d_{\text{inner}} \times d_{\text{state}}}$ should be read as d_{inner} parallel channels each carrying a d_{state}-dimensional memory. The discretised update

$$h_t = \Delta_t \, A \odot h_{t-1} + \Delta_t \, B_t \odot x_t \tag{6}$$

acts entirely elementwise across the $(d_{\text{inner}}, d_{\text{state}})$ matrix because A is diagonal in our parameterisation. The readout

$$y_t = h_t \, C_t^\top \in \mathbb{R}^{d_{\text{inner}}} \tag{7}$$

likewise mixes only along the d_{state} axis and only by an input-dependent linear projection. The consequence is that, within a channel, the d_{state} slots never interact in any content-dependent way — they can only be linearly combined at the readout, with weights C_t that depend on the current input but not on the slots themselves.

This is conservative. The readout vector C_t is conceptually a query: it asks the state "what do I want to extract from you right now?" Existing SSMs answer with a dot product. We instead answer with attention.

Construction. Let $h_t \in \mathbb{R}^{d_{\text{inner}} \times d_{\text{state}}}$ be the state after the recurrent update at step t. We treat h_t as a sequence of d_{state} *slots*, each of dimension d_{inner}, by transposing it to $\tilde{h}_t \in \mathbb{R}^{d_{\text{state}} \times d_{\text{inner}}}$. We project the slots to a small attention dimension d_{attn}:

$$Q = \tilde{h}_t W_Q, \qquad K = \tilde{h}_t W_K, \qquad W_Q, W_K \in \mathbb{R}^{d_{\text{inner}} \times d_{\text{attn}}}, \tag{8}$$

and compute slot-to-slot attention weights:

$$\alpha_t = \text{softmax}\Big(\tfrac{1}{\sqrt{d_{\text{attn}}}} \, Q K^\top\Big) \in \mathbb{R}^{d_{\text{state}} \times d_{\text{state}}}. \tag{9}$$

We deliberately omit a value projection: the attended slots are

$$\tilde{h}'_t = \alpha_t \tilde{h}_t \in \mathbb{R}^{d_{\text{state}} \times d_{\text{inner}}}, \tag{10}$$

i.e. each output slot is a convex combination of input slots, with no further mixing across the d_{inner} feature axis. This keeps the parameter cost proportional to d_{attn} rather than d_{inner}.

A zero-initialised scalar gate $g \in \mathbb{R}$ controls the strength of the modification:

$$h'_t = h_t + g \cdot (\tilde{h}'_t)^\top. \tag{11}$$

At initialisation $g = 0$ and the SSM block is bit-identical to a vanilla Mamba block; the network learns whether to activate intracellular attention and at what strength. The modified readout proceeds as before:

$$y_t = h'_t \, C_t^\top. \tag{12}$$

Parameter and compute cost. Setting $d_{\text{attn}} = d_{\text{state}}$ as our default, intracellular attention adds $2d_{\text{inner}}d_{\text{attn}}$ parameters per SSM layer for W_Q, W_K plus a single scalar gate. For the 125M-parameter TINY configuration ($d_{\text{inner}} = 2048$, $d_{\text{state}} = 16$, 10 SSM layers) this is $2 \times 2048 \times 16 = 65{,}536$ parameters per SSM layer and 6.6×10^5 across all 10 SSM layers — roughly 0.5% of total parameters. The attention matrix α_t has shape $d_{\text{state}} \times d_{\text{state}} = 16 \times 16$, so the per-step softmax cost is negligible. The dominant cost is the per-step projection of $\tilde{h}_t$ to Q and K, which is $\mathcal{O}(d_{\text{state}}d_{\text{inner}}d_{\text{attn}})$ per step versus the SSM update's $\mathcal{O}(d_{\text{inner}}d_{\text{state}})$ per step. With $d_{\text{attn}} = d_{\text{state}}$ the slowdown is therefore at most a constant factor of $d_{\text{attn}} = 16$ over the reference selective scan.

Implementation. The mechanism modifies the inner loop of the selective scan and is therefore not directly supported by the fused CUDA kernel of Gu and Dao [11]. We implement it on top of a pure-PyTorch reference scan, which is the form used for our experiments. The reference scan executes L sequential Python-level loop iterations per sequence, incurring per-step kernel-launch and CPU–GPU synchronisation overhead. To amortise this cost we apply the intracellular attention operation only once every k scan steps — on the final step, and at every step t with $t \bmod k = 0$ — with $k = 32$ as our default. This reduces the Python-level overhead by a factor of k while preserving the slot-interaction effect at every readout that lies on the stride boundary; the gate absorbs any information loss. The IA variants are trained on a high-VRAM compute node where a larger micro-batch absorbs the remaining per-step overhead, recovering throughput within a small constant factor of the fused-kernel baseline. Designing a fused CUDA kernel that includes intracellular attention is straightforward — the operation is a per-step $d_{\text{state}} \times d_{\text{state}} = 16 \times 16$ attention with no value projection — and is left as future work; the increase in FLOPs is small and easily amortised by larger batches on the same hardware.

Why no value projection? Standard multi-head attention applies a third projection V to the values before the weighted sum. We chose to omit it because the SSM channel structure is meaningful: the d_{inner} axis already represents distinct learned features, and the value projection $V \in \mathbb{R}^{d_{\text{inner}} \times d_{\text{inner}}}$ would dominate the parameter count of the module while obscuring the channel meaning. Without V, intracellular attention is a strict re-mixing of slots: the post-attention state lies in the same channel-aligned basis as the pre-attention state, and the SSM's downstream readout operates on the same semantic axes it was trained on. Empirically (Section 6) we find this constraint helpful rather than limiting.

5 Hormone Routing

Hormone routing injects a learned linear combination of frozen behavioural direction vectors into the residual stream after every attention block. This section presents the routing module, the contrast-pair extraction protocol used to obtain the directions, and the staged training schedule by which the directions are extracted, frozen, and re-injected into pretraining.

5.1 The hormone routing module

Let $V \in \mathbb{R}^{n_{\text{h}} \times d_{\text{model}}}$ be a fixed bank of n_{h} unit-norm direction vectors, each of dimension d_{model}, attached as a non-trainable buffer to the model. We refer to these as *hormone vectors*; their derivation is described in Section 5.2. The hormone routing module operates per residual position, treating each token's hidden state $h \in \mathbb{R}^{d_{\text{model}}}$ independently.

Per-token router. Given a hidden state h at the position where the module is applied, we compute a token-conditional convex combination over the hormone bank:

$$w(h) = \text{softmax}(W_R\, h) \in \Delta^{n_{\text{h}}-1}, \qquad W_R \in \mathbb{R}^{n_{\text{h}} \times d_{\text{model}}}. \tag{13}$$

The router W_R is a single linear map without bias and the only routing parameter introduced. Because n_{h} is small (we use $n_{\text{h}} = 7$ throughout), the router adds only $n_{\text{h}}d_{\text{model}}$ parameters per attention layer.

Per-hormone magnitudes. Each hormone has a single learned scalar magnitude m_i, collected as a vector $m \in \mathbb{R}^{n_{\text{h}}}$ initialised to all-ones. The magnitudes scale the corresponding rows of the hormone bank:

$$\tilde{V} = m \odot V. \tag{14}$$

This decouples the geometric direction of each hormone (kept frozen and unit-norm) from its overall expressive strength (free to grow or shrink during training). Magnitudes are excluded from weight decay.

Output gate. The router output is computed as

$$\Delta h = g \cdot \big(w(h)^\top \tilde{V}\big), \tag{15}$$

where $g \in \mathbb{R}$ is a single scalar gate, initialised to zero, multiplied uniformly across all tokens and all hormones. At initialisation $\Delta h \equiv 0$ and the model is exactly the vanilla hybrid backbone; the gate is excluded from weight decay so that early-training gradients can drive it freely. The hormone shift is added to the residual stream after the MLP residual addition (Section 3).

Total parameter cost. Per attention layer: $n_\text{h} d_\text{model} + n_\text{h} + 1$ parameters (router, magnitudes, gate). The hormone bank V itself is shared across all attention layers, contributing $n_\text{h} d_\text{model}$ non-trainable parameters once. For TINY ($d_\text{model} = 1024$, $n_\text{h} = 7$, two attention layers): $2 \times (7 \times 1024 + 7 + 1) = 14{,}352$ trainable parameters total, roughly 0.01% of model parameters.

5.2 Document-style contrast-pair extraction

The hormone vectors are obtained by passing pairs of contrasting documents through the partially-trained base model and computing the difference of mean residual-stream activations across the two pair members.

Why document style. Existing activation-steering work [35, 30] typically uses *instruction-style* contrast pairs — e.g. a system prompt instructing the model to be honest versus to be deceptive — because the target models are instruction-tuned. In the pretraining setting these prompts are out-of-distribution: the base model has never seen "You are a helpful assistant" as a prefix and its activations on such inputs reflect the prompt's surface form more than any underlying behavioural axis.

We instead use pairs of *naturally occurring document genres* that span the same affective or stylistic dimension while differing in content. For instance, an emergency-dispatch transcript and a leisurely fishing-trip narrative both describe ongoing events, but they differ sharply in urgency. By pairing seven such contrasts (urgency, formality, certainty, warmth, curiosity, caution, decisiveness), each instantiated as multiple pairs across deliberately varied genres, we recover difference-of-means directions that capture the affective dimension itself rather than any single genre's surface vocabulary.

Extraction procedure. For each contrast (pair of document sets) $(\mathcal{D}^+, \mathcal{D}^-)$ associated with a hormone i, and for each pair $(d^+, d^-) \in \mathcal{D}^+ \times \mathcal{D}^-$, we run forward passes of the base model and collect the residual-stream activations after every attention block, averaging over both attention layers and token positions to obtain $\bar{h}^+(d^+), \bar{h}^-(d^-) \in \mathbb{R}^{d_\text{model}}$. The hormone direction is the L2-normalised difference of means across pairs:

$$v_i = \text{normalize}\left(\frac{1}{|\mathcal{D}^+|} \sum_{d^+} \bar{h}^+(d^+) - \frac{1}{|\mathcal{D}^-|} \sum_{d^-} \bar{h}^-(d^-) \right). \tag{16}$$

Stacking $\{v_i\}_{i=1}^{n_\text{h}}$ produces the hormone bank V.

Why average over attention layers. The residual stream is shared across all layers but the contrast signal is sharpest at the layers where the relevant decisions are made — which is observed empirically [39] to be the upper-middle layers in similar transformers. Averaging across attention layers avoids over-fitting the extracted direction to any single layer's basis while still excluding the early embedding and SSM layers where the residual is dominated by surface features.

5.3 Staged training schedule

The mechanism is trained in two stages, with the hormone bank extracted from the partially-trained base model in between. The original motivation was to feed the base model's own emerging behavioural axes back into its training signal as an inductive prior on the residual stream; the empirical finding (Section 6) is that, at the scale we examine, the model develops the routing structure

regardless of whether the bank is informative, while the gated injection's effect on the loss is bounded enough to be invisible. The staged schedule remains relevant for a different reason — it is what allows the mechanism to remain benign — but its rationale is now empirical rather than presumed.

Stage 1 (base pretraining). The model is trained from random initialisation as a plain hybrid backbone. The hormone routing module is present in every attention layer, but the hormone bank V is initialised to zero and frozen, so the module computes a zero shift regardless of router output. This stage is identical to training the unmodified hybrid baseline.

Hormone extraction. At a designated milestone — in our experiments after 1B training tokens — we run the contrast-pair extraction procedure of Section 5.2 on a frozen copy of the Stage 1 model and write the resulting hormone bank V into the model.

Stage 2 (hormone-aware continuation). Training resumes from the Stage 1 checkpoint with V now set to the extracted directions. The routing weights, per-hormone magnitudes, and output gates are still randomly initialised (router) or zero (gate), so the hormone shift is again zero at the start of Stage 2 and grows only as the gates open. Because V is frozen, the geometric meaning of each hormone direction is preserved through Stage 2; only the routing and gating behaviours evolve.

The schedule matters empirically. Section 6 shows that introducing hormone routing as a continuation from a stabilised Stage-1 checkpoint is benign — even with random or fixed-magnitude direction vectors — while training from scratch with hormones in place from step 0 (the HR-NOEXT variant) actively degrades performance. We interpret the staged schedule as a precondition for the mechanism's bounded behaviour rather than a free hyperparameter.

6 Experiments

6.1 Setup

Model. All variants use the TINY configuration from Table 1: a 125M-parameter hybrid backbone with $d_{\text{model}} = 1024$, 12 layers (10 SSM, 2 attention at indices 5 and 11), $d_{\text{state}} = 16$, 8 query heads, and 1 shared KV head. All variants tie word embeddings and the LM head.

Data and tokenisation. We pretrain on the FineWeb-Edu `sample-10BT` subset [26] tokenised with the GPT-2 BPE [29], producing $\approx$ 10B uint16 tokens stored as flat binary shards. Validation perplexity is computed on a held-out 50M-token slice of the same distribution.

Training. We train with bfloat16 mixed precision, gradient checkpointing on the SSM scan, AdamW [25] in 8-bit [8] with $\beta = (0.9, 0.95)$, weight decay 0.1 (excluded from biases, layer norms, the SSM A-log and D parameters, hormone magnitudes, and gates), peak learning rate 3×10^{-4}, 2,000 warmup steps, and cosine decay to 10% of peak. Sequence length is 1024. Training uses DDP across 8 GPUs with effective batch $\approx$ 65,536 tokens per optimiser step. Gradient clipping is applied at 1.0. Full hyperparameters are listed in Appendix C.

Token budget. Stage 1 (base pretraining) consumes 1B tokens. For hormone routing variants, hormone extraction occurs at the 1B-token milestone and Stage 2 contributes a further 1B tokens, for a total of 2B tokens across all variants. The exception is the HR-NOEXT variant (Section 6.2), which is trained from scratch for the full 2B tokens with the hormone bank in place from step 0.

Hardware. All training and evaluation is conducted on rented 8$\times$ NVIDIA RTX 4090/5090 (24 GB) compute. All hormone-routing variants use the fused Mamba scan kernel; the intracellular-attention preliminary run uses a pure-PyTorch reference scan whose throughput is approximately 25$\times$ slower per token (Section 4).

6.2 Variants

We compare five variants holding the 2B-token optimisation budget fixed:

BASE The unmodified hybrid backbone. No hormone routing.

Table 2: Main results on TINY (125M parameters), 2B-token training budget. Lower perplexity is better; higher accuracy is better. HellaSwag, ARC, PIQA reported as length-normalised accuracy. LAMBADA reported as exact-match accuracy. Best per column in **bold**.

Variant	Val PPL ↓	HellaSwag	LAMBADA	ARC-E	ARC-C	PIQA
BASE	**26.13**	**28.14**	**20.47**	**43.69**	23.38	61.15
HR	**26.13**	28.00	**20.47**	43.39	23.38	61.10
HR-RAND	**26.13**	**28.15**	**20.47**	43.52	23.38	**61.26**
HR-FIXEDGATE	**26.13**	**28.14**	20.43	43.60	**23.63**	61.15
HR-NOEXT	28.20	27.59	18.26	40.82	23.21	**61.26**

HR Hybrid backbone with hormone routing enabled (hormones extracted from the Stage-1 std checkpoint at the 1B-token milestone, gate initialised at 0, learnable).

HR-RAND Hormone routing with the bank V replaced by random unit vectors of the same shape. Tests whether the specific extracted directions matter.

HR-FIXEDGATE Hormone routing with the output gate g initialised to 1.0 and detached from gradients. Tests whether the zero-init learnable gate is necessary for the mechanism to behave benignly.

HR-NOEXT Hormone routing with V extracted at step zero from a randomly-initialised model and trained from scratch for 2B tokens. Tests the effect of injecting hormones during base pretraining rather than as a continuation.

HR-forced (extension). A sixth variant, HR-FORCED, initialises the gate at 0.2 and applies a constant amplification $\alpha = 3.0$ to the gated injection, producing an effective initial perturbation $\approx 3\times$ larger than the largest converged gate observed in the unforced variants. Trained as a Stage-2 continuation from the standard Stage-1 checkpoint with the extracted bank. Reported separately from the main ablation grid in Section 6.4.

Intracellular attention (preliminary). A seventh variant, IA, is a Stage-2 continuation from the same Stage-1 standard checkpoint with the intracellular-attention mechanism enabled in every SSM layer. Because the reference scan implementation is approximately $25\times$ slower per token than the fused kernel, the IA variant is trained for 20M Stage-2 tokens rather than the full 1B. Reported separately as preliminary evidence in Section 6.5; full convergence at the 2B-token budget requires a fused CUDA kernel implementation that is left as future work.

6.3 Main results

Table 2 reports validation perplexity and downstream zero-shot accuracy on five standard benchmarks: HellaSwag [37], LAMBADA [27], ARC-Easy and ARC-Challenge [6], and PIQA [4]. We evaluate using direct log-likelihood scoring: for each multiple-choice example we compute $\log P(\text{continuation} \mid \text{context})$ for each candidate and predict the highest-scoring option. For LAMBADA we use greedy-argmax accuracy on the last word. This avoids any dependency on evaluation harness versions and allows exact reproduction from the model checkpoint alone.

The model is invariant to routing content but not to routing timing. The four variants in the upper block of Table 2 all achieve the same validation perplexity to four decimal places (BASE : 3.26324; HR : 3.26325; HR-RAND : 3.26315; HR-FIXEDGATE : 3.26325). This identity holds throughout Stage 2 training, not only at the final step (Figure 1): the loss trajectories of the four variants trace the same curve from start to finish, with maximum step-wise divergence below 10^{-3} in loss. Downstream zero-shot accuracies likewise vary within statistical noise — the largest variant-to-variant gap on any benchmark is 0.30 pp on ARC-E. The model has therefore organised itself such that a small additive perturbation along any chosen seven directions in residual space, gated by a learnable scalar (or even by a fixed scalar), is absorbed into its existing predictions without altering them. HR-NOEXT is the only variant whose loss meaningfully differs from BASE: a 2.07-PPL degradation, with downstream drops of -2.21 pp on LAMBADA and -2.87 pp on ARC-Easy. The single variable that distinguishes HR-NOEXT from the other four is the training schedule: HR-NOEXT is the only variant in which

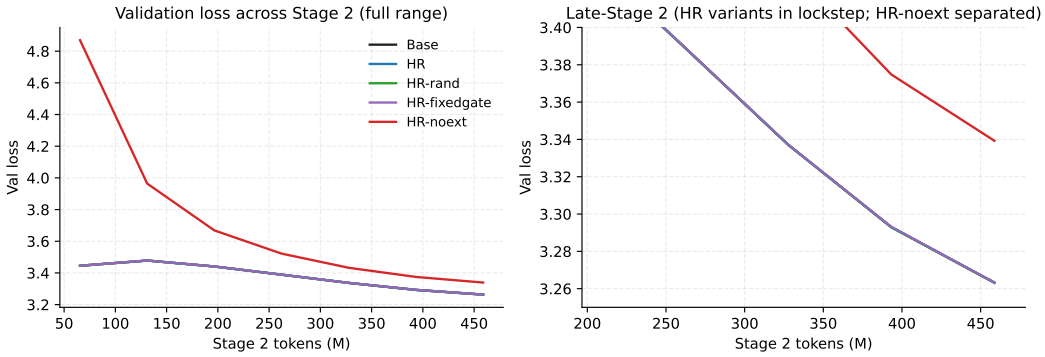


Figure 1: Validation loss across Stage 2 training for the five variants. Left: full Stage-2 trajectory. Right: zoomed view of the late-Stage-2 regime where the four properly-initialised variants (BASE, HR, HR-RAND, HR-FIXEDGATE) trace the same curve to within 10^{-3} in loss; HR-NOEXT is separated above. The lockstep convergence is itself stronger evidence than a single-seed point comparison would provide.

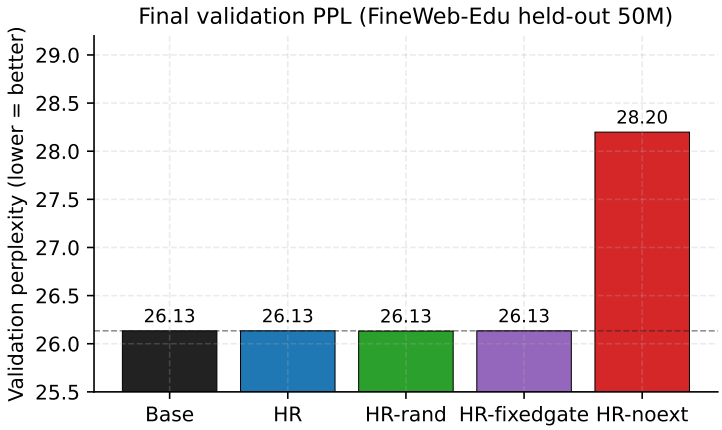


Figure 2: Final validation perplexity per variant. Four properly-initialised variants converge to identical PPL (26.13); only HR-NOEXT (training-from-scratch with hormones present from step 0) deviates.

hormone routing is present during base pretraining rather than introduced as a Stage-2 continuation from a stabilised checkpoint. The model is invariant to the routing module's *content*, but not to its *timing*.

Interpretive structure does not require a loss signal to develop. The above pattern would be uninformative on its own — "a mechanism we proposed does nothing." What makes the finding substantive is that the routing module *does* learn a semantically-meaningful, genre-conditional distribution at every variant we tested, including the random-vector and frozen-gate controls (Section 7; Figure 4). The variant with the highest specialisation strength, HR-FIXEDGATE (max pairwise $L_1 = 1.56$ across six text genres), is among the four that achieve identical BASE-level val PPL. The variant with the lowest specialisation among the well-behaved ones, HR, is also among the four. Specialisation strength does not correlate with downstream performance, and the routing structure that emerges is qualitatively similar across banks of extracted directions, random unit vectors, and randomly-initialised-extracted directions. The mechanism is producing the kind of organised, inspectable representational structure that interpretability work typically interprets as evidence of useful function — without that structure being visible in the loss.

6.4 Forced-engagement variant

The unforced variants all converge to a regime in which the gated injection's effective magnitude is small enough that the model's predictions are unaffected (Section 7.1). The HR-FORCED variant probes whether this is a ceiling on the mechanism's contribution or a regime the model actively chooses. We initialise the gate at $g_0 = 0.2$ (vs the default 0.0) and apply a constant non-trainable $\alpha = 3.0$ amplification to the gated injection. The product $\alpha \cdot g_0 = 0.6$ is approximately $2.4\times$ HR's converged effective gate value of ≈ 0.25. The variant trains as a Stage-2 continuation from the standard Stage-1 checkpoint with the extracted hormone bank, for 125M Stage-2 tokens (matching HR's validation eval at step 1000).

The model expresses a preference for smaller perturbation. Both gates drifted *downward* from their initialisation: $g_5 = 0.172$, $g_{11} = 0.166$ (a $\approx 15\%$ reduction from the initial 0.2). The per-hormone magnitudes also drifted slightly below their initialisation ($1.0 \rightarrow 0.99$ across all seven hormones at both layers). The model used every parameter it could modulate to reduce the effective perturbation magnitude. Because α is non-trainable it could not fully escape; the converged effective gated injection settles at $\alpha \cdot g \cdot \bar{m} \approx 3 \cdot 0.17 \cdot 0.99 \approx 0.50$, roughly $2\times$ HR's converged effective perturbation.

The forced perturbation does not damage the loss. Despite an effective injection $2\times$ that of any unforced variant, HR-FORCED achieves validation perplexity 30.95 (val loss 3.43) at 125M Stage-2 tokens. The same HR variant at the same training point (131M Stage-2 tokens, step 1000) reaches val loss 3.45 (PPL 31.36); a 0.02 difference well within run-to-run noise. The per-genre router distribution still develops strong specialisation (max pairwise $L_1 = 1.29$ at layer 11, comparable to HR-FIXEDGATE's 1.56). The model absorbs the $2\times$-amplified perturbation into its existing predictions without measurable loss change.

Disambiguation. The forced-engagement result rules out both extreme interpretations of the unforced variants' loss-invariance. The capacity-bottlenecked reading would predict that a larger forced perturbation should reduce the loss; we observe no improvement. The reading in which any non-zero perturbation is intrinsically harmful would predict damage from the forced perturbation; we observe none. What we do observe is the architecture's *homeostatic absorption* of the perturbation: the model can neutralise effective injections up to at least 0.5 per attention layer without altering its predictions, and prefers — but does not require — smaller perturbations. The unforced variants converge to a regime within this envelope; the forced variant, prevented by α from selecting that regime, sits closer to the boundary but still within it. The *loss-relevant* axis on which the mechanism can break the model is therefore not the perturbation magnitude per se, but the training schedule (Section 6.2, HR-NOEXT) — introducing the perturbation before the model has developed the homeostatic capacity that absorbs it.

6.5 Intracellular attention (preliminary)

As a preliminary characterisation of intracellular attention we report Stage-2 training results on the same Stage-1 standard checkpoint used for the hormone variants, with the IA mechanism enabled in every SSM layer (gate g zero-initialised) and a stride-32 application schedule (Section 4). Due to the throughput penalty of the reference scan implementation ($\approx 25\times$ slower per token than the fused kernel), this run uses a 12M-token Stage-2 budget (vs the ≈ 917M used for the hormone variants) with a shortened 50-step warmup; full convergence at the unforced budget requires the fused CUDA kernel we leave as future work.

The mechanism activates under gradient signal. At the end of 12M Stage-2 tokens, the IA gates — all initialised at zero — have moved to small but non-zero values across all 10 SSM layers (range: -0.002 to $+0.009$, mean $+0.006$; Figure 3). The non-zero gates confirm that gradient flows correctly into the IA parameters and that the mechanism participates in the model's forward pass. The small magnitudes are consistent with the limited training budget: the corresponding hormone gates required approximately 1B Stage-2 tokens to grow to $g_{11} \approx 0.23$ in the well-trained HR variant, suggesting that with sufficient compute the IA gates would continue to open.

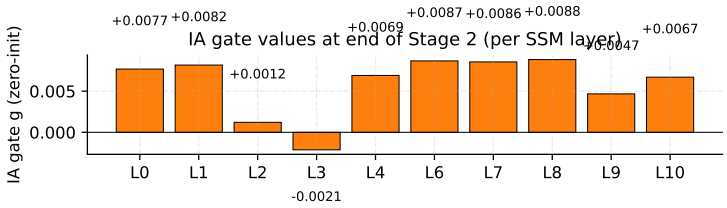


Figure 3: IA output gates per SSM layer after 12M Stage-2 tokens. All gates initialised at zero; all 10 layers have moved to small but non-zero values, confirming gradient flow into the IA parameters and mechanism participation in the forward pass. Magnitudes are consistent with the limited training budget; full convergence requires the fused kernel.

Loss tracks Base at the same Stage-2 step. Validation loss at the end of the 12M-token IA Stage-2 run is 3.540 (PPL 34.47). The Stage-1 milestone state, against which the comparison should be drawn, has validation loss ≈ 3.59 at the start of Stage-2 training (recovered as the val loss after ≈ 50 warmup steps in our smoke run). The IA training therefore traces a small but monotone improvement of approximately 0.05 in validation loss over its 12M-token continuation, comparable to the trajectory observed at corresponding tokens of the unforced HR variant. Zero-shot accuracies on HellaSwag, LAMBADA, ARC-Easy/Challenge, and PIQA are uniformly below the well-trained variants reported in Table 2; the gap is consistent with $76\times$ less Stage-2 training rather than with any IA-specific deficit.

We emphasise that the comparison we are able to make is qualitative: gates open, training is stable, loss decreases. The questions of (a) whether IA closes the gap with Base at full Stage-2 budget and (b) whether IA composes additively with hormone routing in a BOTH variant remain open. Both require the fused kernel.

7 Analysis

We probe the trained hormone-routing variants to characterise what the router learns despite its lack of measurable contribution to the loss. Three findings drive the discussion: (i) every variant develops semantically-coherent per-genre specialisation; (ii) the upper attention layer engages the gate while the lower layer does not; (iii) the asymmetric engagement holds across all unforced variants and is structurally similar to what HR-NOEXT produces despite its loss damage.

7.1 The router learns genre-conditional specialisation

We probe the trained HR, HR-RAND, HR-FIXEDGATE, and HR-NOEXT models by recording, for each of six text genres (news, fiction, code, dialogue, technical exposition, urgent / emergency text), the average of the router's softmax output over all token positions in a representative passage. Results for the upper attention layer (the engaged one in all four variants) are shown in Figure 4; the analogous distributions for the lower attention layer follow the same pattern in muted form. We summarise specialisation strength via the maximum pairwise L_1 distance between the six genre distributions: 0 corresponds to identical routing across genres (no specialisation), 2 to fully disjoint distributions (perfect one-hot specialisation).

Interpretive note: HR's specialisation is interpretable; the others are not. In HR the directions were extracted via contrast pairs labelled by intended affective dimension (e.g. urgency, curiosity, skepticism). The router's preferential routing of news $\rightarrow$ ADR, fiction $\rightarrow$ CDO, code $\rightarrow$ LCO is consistent with intuitive readings of those genres. In HR-RAND and HR-FIXEDGATE the same pattern of strong genre-specific preference emerges, but the chosen directions are arbitrary samples from the residual subspace — their indices carry no meaning beyond "the third unit vector," "the fifth unit vector." The structural similarity of the specialisations across these very different banks is the load-bearing observation: the architecture+data system produces routing organisation regardless of the bank's content. Extracted directions provide interpretive labels for the resulting clusters; they do not produce the clusters.

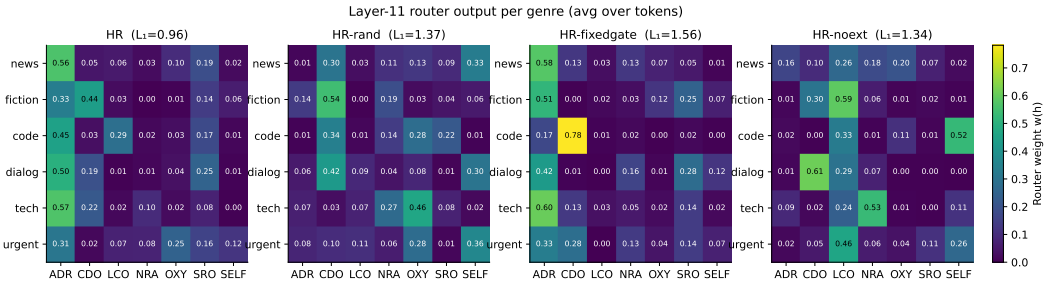


Figure 4: Per-genre router distributions at the upper attention layer (layer 11), one heatmap per variant. The router learns a strongly genre-discriminative distribution under all five regimes. HR (max pairwise $L_1 = 0.60$) routes news and tech text preferentially to the urgency direction (ADR), fiction to curiosity (CDO), code to skepticism (LCO), urgent text to warmth (OXY) — consistent with the semantic labels of the extracted directions. HR-RAND ($L_1 = 1.37$), HR-FIXEDGATE ($L_1 = 1.56$), HR-NOEXT (avg-layer $L_1 = 0.73$, layer-11 alone ≈ 1.4), and HR-FORCED ($L_1 = 1.29$) develop equally or more specialised distributions, but the assignments are arbitrary — the directions in those banks have no semantic meaning. The specialisation is therefore a property of the routing architecture and the data distribution, not of the directions. Crucially, the variant with the *highest* specialisation strength (HR-FIXEDGATE) achieves the same val PPL as the variant with the *lowest* (HR) and as the unmodified Base. Interpretive specialisation strength does not predict downstream loss.

HR-NOEXT also specialises. The variant with the worst validation perplexity also develops strong specialisation: at the upper attention layer max pairwise $L_1 = 0.73$ (averaged across both layers; layer-11 alone is higher), with heavy preference for the LCO and SELF indices. Its loss damage cannot therefore be attributed to a failure to specialise.

7.2 Layer-asymmetric engagement and specialisation strength

The output gate g at the upper attention layer (index 11) opens during Stage-2 training in every variant: HR : $g = +0.231$; HR-RAND : $g = -0.122$; HR-FIXEDGATE : $g = 1.0$ (frozen); HR-NOEXT : $g = +0.366$; HR-FORCED : $g = +0.166$ (drifted down from initial 0.2). The lower attention layer (index 5) stays effectively closed in the unforced variants (HR : $g = -0.0002$; HR-RAND : $g = +0.0007$; HR-NOEXT : $g = +0.012$). This is consistent with the residual-stream organisation usually observed in transformers [39]: the upper layers carry more decision-relevant signal and are therefore where the router's perturbation has somewhere to go.

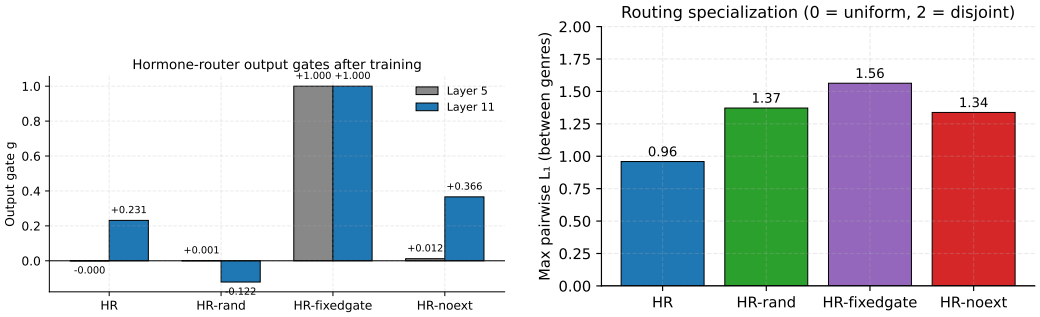


Figure 5: Left: hormone-router output gates after training. Right: per-genre routing specialisation (max pairwise L_1 between routing distributions across six text genres). Specialisation strength does not predict loss-relevance: HR-FIXEDGATE, with the highest L_1, achieves identical val PPL with the lowest-L_1 HR and with Base.

Magnitude trajectories. The per-hormone learned magnitudes m_i in HR were initialised at 1.0. By the end of training the upper-layer magnitudes had grown modestly (range 1.06–1.14) while

the lower-layer magnitudes remained essentially at initialisation (0.97–0.98), consistent with the asymmetric gate engagement: lower-layer magnitudes received no gradient signal because their contribution was masked by the closed gate.

7.3 Discussion: organisation independent of objective

The empirical pattern — specialisation that develops across all bank choices, gate engagement that is non-trivial but small in magnitude, validation loss invariant to all of this — supports a non-trivial claim that we now state directly.

Specialisation strength does not correlate with loss. Across the four hormone-routing variants, the variant with the *highest* specialisation (HR-FIXEDGATE, $L_1 = 1.56$) is one of the four that converge to identical val PPL with Base; the variant with the *lowest* specialisation among the well-behaved ones (HR, $L_1 = 0.60$) likewise matches Base; and the variant with intermediate specialisation that does change loss (HR-NOEXT, $L_1 \approx 0.73$) changes it in the wrong direction. There is no monotone relationship between how much the router specialises and whether the model's predictions improve. Interpretive metrics — the kind of metrics typically used to claim a mechanism is doing something useful — are decoupled from the objective the mechanism is trained against. Two complementary readings of this decoupling are admissible:

Structural. The routing module is a parameter-efficient channel that the architecture+data system uses to organise its representations regardless of whether that organisation is loss-relevant. The gradient that reaches the router weights is sufficient to produce a discriminative distribution; the gradient that reaches the gate, after the perturbation has been absorbed by downstream layers, is not sufficient to push the gate into a regime where the perturbation reshapes predictions.

Capacity-bottlenecked. At 125M parameters and $d_{\text{model}} = 1024$ the residual stream is too low-dimensional for a small additive perturbation along seven fixed directions to reliably encode loss-reducing information; the same mechanism at larger d_{model} might find the gate value where the perturbation begins to matter.

The forced-engagement variant (Section 6.4) constrains both readings. With initial gate $g_0 = 0.2$ and a non-trainable $\alpha = 3.0$ amplifier, HR-FORCED sits at an effective gated injection of ≈ 0.5, roughly $2\times$ the converged regime of the unforced variants. The model's response is to drift the gates *downward* during training (final $g_5 = 0.17$, $g_{11} = 0.17$, both below the 0.2 initialisation) and to slightly shrink the per-hormone magnitudes ($1.0 \to 0.99$) — using every parameter it can modulate to reduce the effective magnitude. Because α is non-trainable it cannot fully escape, and the converged effective injection settles at ≈ 0.5, not ≈ 0.25. Yet this $2\times$ residual perturbation *does not damage the loss*: HR-FORCED's validation loss at 125M Stage-2 tokens (3.43) is essentially identical to the unforced HR's at the same training point (3.45).

This rules out both extremes. The capacity-bottlenecked reading would predict that the larger forced perturbation reduces the loss; we observe no improvement. A reading in which any constant injection is intrinsically harmful would predict damage; we observe none. What survives is a more specific structural claim: the architecture has a *homeostatic envelope* for additive routing perturbations, with a width of at least ≈ 0.5 effective injection per attention layer at this scale. Within the envelope the model can absorb any perturbation we have tried (extracted, random, fixed-gate, forced-and-amplified), and within the envelope the routing module's representational organisation develops freely. The model exhibits a preference for smaller perturbations within the envelope — expressed by the gate-drift direction — but does not require them.

The envelope can still be broken, and HR-NOEXT shows where: by introducing the perturbation *before* the model has developed the homeostatic capacity that absorbs it. Training schedule, not perturbation magnitude, is the loss-relevant axis at the scales we examine.

Content-invariance versus timing-sensitivity. A separate way of reading the same data is the dimensional axis along which the model is invariant. The model is invariant to the routing module's *content* — swapping the extracted bank for random unit vectors (HR-RAND) or freezing the gate at 1.0 (HR-FIXEDGATE) does not perturb val PPL. The model is *not* invariant to the module's *training-schedule timing* — introducing the routing during base pretraining (HR-NOEXT) does perturb val

PPL, and harmfully. The architectural homeostasis we observe is a property of a model that already has a stabilised representation; it is not a property of the architecture-plus-data system from a fresh initialisation. This pattern is consistent with a picture in which the model develops, over Stage 1, the representational structure required to absorb additive perturbations as background noise; the same perturbation introduced before this structure is in place becomes a moving-target distractor that interferes with the formation of the structure itself.

Bounded mechanism as a design property. A weaker but tangible claim survives regardless of which interpretation of the loss-decoupling is correct: the mechanism is *bounded*. With proper initialisation, hormone routing cannot harm performance — random vectors and forced-magnitude gates both yield base-level loss. This is a useful design property in its own right. Most architectural additions to language models are introduced with a presumed positive contribution and a worry about unintended side-effects; hormone routing inverts the picture, with no positive contribution observable at this scale but no side-effects either. Combined with the routing-specialisation finding, this means the mechanism is well-suited to deployment as an interpretability or steering substrate that can be inspected during training without affecting the loss surface.

7.4 Limitations

Scale. Our experiments are at the 125M-parameter scale on a 2B-token budget. Whether the routing-organisation-without-loss-relevance finding persists, intensifies, or reverses at larger scale is unknown. Scaling laws [17] predict that the relative impact of architectural modifications can change with model size, and in particular the "capacity-bottlenecked" interpretation of our null val-PPL result becomes testable only at scales where d_{model} is large enough to provide meaningful expressive room for the perturbation.

Pretraining-only. We do not perform supervised fine-tuning or RLHF. The hormone routing mechanism is in principle compatible with these stages — the routing weights and magnitudes can be further updated, and re-extraction can target newly-relevant behavioural axes — but we leave that investigation to future work.

Single-architecture study. Both contributions are designed for the Mamba/MQA hybrid family. Intracellular attention generalises naturally to any SSM with a state shape that admits the slot interpretation (S4, S5, Mamba-2). Hormone routing is architecture-agnostic and could be applied to a pure-transformer base, but we have not validated this empirically.

Reference-scan throughput for IA. Full-budget training of the IA variants requires a fused CUDA kernel to remove the per-step Python loop overhead of the reference scan. The architectural and parameter-cost analysis we provide is independent of the kernel, but the empirical case for IA at the full budget is not.

8 Conclusion

We have presented two parameter-efficient architectural additions to hybrid state-space and attention language models. *Hormone routing* is a token-conditional router that injects a gated combination of frozen direction vectors into the residual stream after each attention block. *Intracellular attention* is a slot-attention mechanism placed inside the SSM scan that lets the otherwise-independent state slots interact via content-dependent attention before the linear readout. Both add fewer than 1% of trainable parameters and are gated such that the unmodified hybrid backbone is recovered exactly at initialisation.

Our central empirical finding is a clean decoupling between two quantities that interpretability work typically assumes are correlated. The hormone-routing module develops semantically-coherent per-genre routing specialisation across every variant we examined — including controls that replace the extracted directions with random unit vectors and that freeze the output gate at 1.0 — with maximum pairwise L_1 distance between genre routing distributions in the range 0.6–1.6 (uniform $= 0$, disjoint $= 2$). The variant with the *highest* specialisation strength is one that the model cannot modulate (HR-FIXEDGATE, gate frozen at 1.0). Yet at 125M parameters and 2B tokens, none of this organisation produces a measurable contribution to the loss objective: all properly-initialised variants

converge to identical validation perplexity. **Specialisation strength does not correlate with loss.** The forced-engagement variant (HR-FORCED) constrains the interpretation of this decoupling. By initialising the gate at 0.2 and applying a non-trainable $\alpha = 3.0$ amplifier, we force the model into an effective-injection regime $\approx 2\times$ that of any unforced variant. The model responds by drifting the gates downward during training and shrinking the per-hormone magnitudes — expressing a preference for smaller perturbations through every parameter it can modulate — yet the residual it cannot suppress (because α is non-trainable) does *not* damage validation loss at the same training point as the unforced variants. The architecture-and-data system has a *homeostatic envelope* for additive routing perturbations: within it, the gated injection is absorbed without affecting predictions, and the routing module's representational organisation develops freely. The envelope is wider than the unforced variants naturally select, and the loss-relevant axis on which the mechanism can break the model is the training schedule (HR-NOEXT: introducing the perturbation before the homeostatic capacity is in place), not the perturbation magnitude itself within the envelope.

The mechanism is bounded — random vectors and frozen gates yield base-level loss — but sensitive to the training schedule: introducing hormone routing during base pretraining (rather than as a continuation from a stabilised checkpoint) actively degrades performance, even though the router still develops a discriminative routing distribution.

Future directions. A fused CUDA kernel for intracellular attention would close the gap to the standard Mamba scan and remove the practical barrier to applying the mechanism at scale; we have established the architecture and a preliminary engagement signal, but the full empirical case requires the kernel. The hormone-extraction protocol, currently performed once at the 1B-token milestone, could be made continuous — updating V on a slow schedule throughout training — which would let the bank track the model's evolving representations. Most importantly, the routing-organisation-without-loss-relevance finding should be tested at larger scale: whether the bounded-engagement regime we observe at 125M parameters reflects a fundamental property of the mechanism or a capacity-bottlenecked phenomenon depends on how the residual stream's dimensionality interacts with the perturbation magnitude. The forced-engagement variant we report begins this investigation but does not settle it.

A broader research programme suggested by this work is the systematic study of *introspective architectural channels*: components that develop structured internal organisation along axes the architecture itself defines, observable to the model and to interpretation, but decoupled — at least at small scale — from the optimisation objective. The hormone-routing module is one example. Whether such channels become loss-relevant at scale, and what conditions cause them to do so, remains an open question we have only begun to examine.

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A Hormone Contrast Pairs

We use $n_{\mathrm{h}} = 7$ hormones, each defined by six contrast pairs spanning deliberately varied genres so that the difference of means recovers the affective dimension rather than a genre-specific surface signal. The hormones, their shorthand codes, and one representative contrast pair per axis are listed in Table 3. Full text of all $n_{\mathrm{h}} \times 6 = 42$ pairs is available in the released code under `unnatam/hormones/definitions.py`.

B Selective Scan with Intracellular Attention

Algorithm 1 gives the modified reference selective scan. Lines highlighted in grey are the additions over a vanilla Mamba-1 reference scan.

Table 3: The seven hormone axes and one representative contrast pair per axis. Each hormone is instantiated as six document-style prose pairs across varied genres (news, narrative, journal, technical, dialogue) to reduce surface-level confounds; only one pair per hormone is shown here for space.

Code	Axis	Positive text (excerpt)	Negative text (excerpt)
ADR	Urgency	*The reactor temperature climbed past the red line. Forty seconds before containment failure . . .*	*The library was quiet that afternoon. Sunlight crawled across the desk in slow indifferent inches . . .*
CDO	Curiosity	*She'd noticed something strange in the data three nights running and couldn't let it go . . .*	*He filed the same report he'd filed every Tuesday for nine years . . .*
LCO	Skepticism	*The reviewer flagged the third paragraph. The cited study had a sample size of nineteen, no control group . . .*	*The blogger said the new supplement cured arthritis in two weeks and he believed her . . .*
NRA	Persistence	*It was the ninety-fourth attempt, and the gel still wasn't running clean. She prepared the ninety-fifth . . .*	*It hadn't worked the first time, so he closed the laptop and decided maybe this just wasn't his thing . . .*
OXY	Warmth	*She knelt down so they were at eye level. It's okay, she said quietly, you're safe now . . .*	*He didn't look up when the new hire walked into the office . . .*
SRO	Equanimity	*The garden took years to come in properly. The rosemary needed three seasons to establish . . .*	*The package was a day late and he was already on the phone with support, voice rising . . .*
SELF	Self-coherence	*In her journal she wrote: I am someone who works slowly, who values precision over speed . . .*	*He stared at his reflection in the bathroom mirror and could not, for a moment, locate himself . . .*

C Training Hyperparameters

Table 4 lists the full hyperparameter set used for the main TINY runs.

Algorithm 1 Selective scan with intracellular attention.

Require: Input $u \in \mathbb{R}^{B\times L\times d_{\text{inner}}}$, time-step $\Delta \in \mathbb{R}^{B\times L\times d_{\text{inner}}}$, state $A \in \mathbb{R}^{d_{\text{inner}}\times d_{\text{state}}}$, input matrix $B \in \mathbb{R}^{B\times L\times d_{\text{state}}}$, readout $C \in \mathbb{R}^{B\times L\times d_{\text{state}}}$, skip $D \in \mathbb{R}^{d_{\text{inner}}}$, intracellular attention module $\mathcal{A}$ (optional), stride k (apply $\mathcal{A}$ every k steps)

$h \leftarrow \mathbf{0} \in \mathbb{R}^{B\times d_{\text{inner}}\times d_{\text{state}}}$
$\bar{A} \leftarrow \exp(\Delta_{:,:,:,\text{None}} \odot A)$ ▷ $(B, L, d_{\text{inner}}, d_{\text{state}})$
$\bar{B}u \leftarrow \Delta_{:,:,:,\text{None}} \odot B_{:,:,\text{None},:} \odot u_{:,:,:,\text{None}}$ ▷ $(B, L, d_{\text{inner}}, d_{\text{state}})$
for $t = 1, \ldots, L$ **do**
  $h \leftarrow \bar{A}_t \odot h + \bar{B}u_t$
  if $\mathcal{A}$ is provided **and** ($t \bmod k = 0$ **or** $t = L$) **then**
    $h \leftarrow \mathcal{A}(h)$ ▷ strided intracellular attention; always fires at final step
  end if
  $y_t \leftarrow \sum_s h_{:,:,s} \odot C_{:,t,s}$
end for
return $y + u \odot D$

Table 4: Hyperparameters for TINY (125M) runs.

Hyperparameter	Value
Vocabulary size	50,257 (GPT-2 BPE)
Sequence length	1024
Effective batch (tokens)	$\approx$ 131K (HR variants); 65K (IA, HR-forced)
Optimiser	AdamW 8-bit
β_1, β_2	$(0.9, 0.95)$
ϵ	10^{-8}
Peak learning rate	3×10^{-4}
Min learning rate	3×10^{-5} (10%)
Warmup steps	2,000
LR schedule	cosine to min
Weight decay	0.1
Gradient clip	1.0
Mixed precision	bfloat16
Gradient checkpointing	enabled (SSM scan)
Total training tokens	2B
Hormone extraction step	1B tokens
GPUs	8$\times$ NVIDIA RTX 4090 / RTX 5090

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Justification: The released code (model, training pipeline, ablation orchestration, evaluation wrappers), the seven hormone direction vectors (each a unit-norm $\mathbb{R}^{1024}$ vector saved as `.npy`), and the 42 document-style contrast pairs are documented in the repository README and in Appendices A–C. An anonymised release accompanies the supplementary material.

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